INCH-POUND

MIL-F-81177E
AMENDMENT 3
21 DECEMBER 1999
SUPERSEDING
AMENDMENT 2
7 DECEMBER 1998

MILITARY SPECIFICATION

FASTENER, BLIND, HIGH STRENGTH, INSTALLATION FORMED, ALLOY STEEL GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-F-81177, dated 15 February 1984, and is approved for use by all departments and agencies of the Department of Defense

PAGE 1

Change paragraph 2.1 to the following:

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Replace paragraph 2.1.1 with the following:

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

PAGE 2

Change the headings "MILITARY" to read "DEPARTMENT OF DEFENSE"

Remove MIL-I-6866 and title.

Remove MIL-I-6868 and title.

Remove MIL-L-8937 and title.

Remove FED-STD-151 and title.

Remove MIL-STD-105 and title.

Remove MIL-STD-1312 and title.

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PAGE 2 (Continued)

Add the following specification under Department of Defense:

MIL-L-46010 Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting

Change the paragraph "(Copies of specifications, standards, drawings....." to:

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, 700 Robbins Avenue, building 4D, Philadelphia, PA 19111-5094.)

PAGE 3

Change paragraph 2.2 to the following:

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are those of the documents cited in the solicitation (see 6.2)

Add the following documents under the American Society for Testing and Materials

ASTM E1417 Standard Practice for Liquid Penetrant Examination

ASTM E144-91 Standard Practice for Magnetic Particle Inspection.

Under American National Standards Institute add the following document:

ANSI/ASQC Z1.4 Sampling Procedures and Tables for Inspection by Attributes

(Copies of American Society for Quality Control documents may be obtained from the American Society for Quality Control, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

Add the following document under the National Aerospace Standards

NASM 1312 Fastener Test Methods

Replace paragraph 2.3 with the following:

2.4 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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PAGE 4

Replace Paragraphs 3.2 and 3.2.1 with the following paragraphs and delete paragraph 3.2.2:

3.2. Verification of Compliance. Manufacturers shall verify that fasteners furnished under this specification meet specified requirements (see 4.3).

3.2.1 Retention of Verification of Compliance. Manufacturers shall maintain on file certified test reports (see 4.3.1) available for inspection by the procuring activity.

PAGE 5

Paragraph 3.3.11: Change the last sentence to read:

Alloy steel serrated pins shall be cadmium plated in accordance with QQ-P-416, Type I or Type II, Class 3.

PAGE 8

Paragraph 3.6.8

Delete the second sentence in its entirety and in the last sentence change MIL-L-8937 to MIL-L-46010.

PAGE 9

Change paragraph 4.2 to the following:

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Verification of Compliance. (see 4.3).

b. Quality Conformance inspection (see 4.4).

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PAGE 10

Change paragraph 4.3 to the following:

4.3 Verification of Compliance. Manufacturers shall verify that fasteners comply with or exceed the following requirements:

Type I - Tensile Preload, Sheet Take-Up, and Tension-Tension Fatigue as specified in paragraphs 3.5.3, 3.5.6 and 3.5.4 respectively.
Type II - Stress Durability, Tension-Tension fatigue, Sheet Take-Up, and Vibration as specified in paragraphs 3.8.8, 3.8.4, 3.8.5, and 3.8.6 respectively.

Change paragraph 4.3.1 to the following:

4.3.1. Certified Test Reports. Manufacturers shall maintain on file certified test reports indicating that both Verification of Compliance and Quality Conformance tests are met. Each report shall be available to the buyer upon request.

Delete paragraph 4.3.2

Delete paragraph 4.3.3

PAGE 11

Paragraph 4.4.2, change MIL-STD-105 to ANSI/ASQC Z1.4.

Paragraph 4.5.2, change MIL-I-6868 to ASTM E1444-91 and change MIL-I-6866 to ASTM E1417.

Paragraph 4.5.3, change MIL-STD-1312 to NASM 1312.

Paragraph 4.5.4, change the first sentence to the following:

Tensile strength tests and tensile preload tests shall be performed in accordance with NASM 1312, Test Methods 8 and 16, except load deflection curves are not required.

PAGE 15

Add 6.2.2:

6.2.2. Acquisition of rivets manufactured prior to issuance of Amendment 1. Unless otherwise specified, Amendment 1 is applicable to rivets manufactured after 16 August 1994."

Delete Paragraph 6.3.

PAGE 16

Renumber paragraphs 6.4 and 6.5 to 6.3 and 6.4.

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PAGE 26

Delete Table VI.

PAGES 27, 28, and 29

Renumber Tables VII, VIII, IX, X, XI, and XII to VI, VII, VIII, IX, X, and XI respectively.

PAGE 30

Delete Table XIII

PAGE 31

Change footnote (a) to the following:

(a) Reqm't Para. 3.3.1, 3.3.4, 3.3.6, 3.3.7, 3.3.8, 3.3.11

PAGES 31, 32, and 33

Renumber Tables XIV and XV to Tables XIII and XIV respectively.

NOTE: Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Appendix A

Tooling Manufacturers

1. Line 2: Delete Cherry Textron and substitute the following "Textron Aerospace Fasteners (TAF), Code Ident. No. 11815"

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Delete Table and Notes, and replace with Table and Notes below.

MS90353 & MS90354	Nominal Size (Diameter)	Huck Mfg. Co. Tool Part No. or equal	Nose Assy. Part No. or equal	TAF Tool Part No. or equal	Nose Assy Part No. or equal	American Pneumatic Tool Nose Assy Part No.
-05 or U05	05/32	200	99-600			
SO5 or U05	05/32	202	99-2701	G83	H83A-5MB*	
		206-375	99-1712	G84	H83A-5MB*	
		206-375	99-1712-1	G704B	H701B-456*	
		244	99-2701	G744	H744-5MB*	
-06 or U06	3/16	200	99-589			
S06 or U06	3/16	206-375	99-1713	G83	H83A-6MB*	
		206-375	99-1713-1	G84	H83A-6MB*	
		202	99-2707	G744	H744-6MB*	
		244	99-2707			
-08 or U08	1/4	206-375	99-1714	G84	H84-8MB*	35050
		245	99-2716	G85D	H652-8*	
		2580	99-2716	G87D	H652-8*	
		5901	99-652			
		2025	99-2713			
		244	99-2713			
-10 or U10	5/16	2580	99-2719			
		5901	H99-599			
		245	99-2719			
-12 or U12	3/8	2580	99-2723			
		5901	H99-599			
		246	99-2723			
-14	7/16	5901	H99-738			
-16	1/2	5901	H99-678			

*** S05, S06, or -08 only.**

-05 and –06 signify Double Action (shifting) Tooling required for 5/32 and 3/16 only.

S05 and S06 signify Single Action (non-shifting) Tooling required for 5/32 and 3/16 only.

-08, -10, -12, -14 and –16 sizes have been and continue to be Single Action (non-shifting) installed.

U code fasteners utilize single action (non-shifting) tooling. For sizes 5/32 and 3/16, double action (shifting) tooling may also be used.

Custodians:
Army - AV
Air Force - 11

Preparing Activity:
DLA-IS
Project no. 5320-1000)

Review Activities:
Army - AR, MI
Navy - AS
Air Force - 99

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